SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

(Deemed to be University)

Department of Mathematics and Computer Science

SYLLABUS: USTA-201: Probability Theory and Distributions

Course Code	USTA-201	Credits: 4 (3-1-0)	Total Periods: 56
Prerequisites	—		

Course Objectives

The main objective of this course is to provide students with the foundations of probabilistic and statistical analysis. The topic covered will include probability theory, random variable and limit theory. The course also focuses on the understanding of discrete and continuous distributions with major emphasis on Bernoulli, Binomial, Poisson, Geometric distributions, Uniform, Cauchy, Normal, Exponential, Gamma distributions.

Course Outcomes

The major outcome from this paper is that the students will be able to explore the basic concepts of probability theory and its applications for decision-making in economics, business, and other fields of social sciences. In addition, the course is oriented towards the formulation of mathematical concepts on probability and probability distributions and densities with practical applications. This enables the students to develop problem-solving techniques needed to accurately calculate probabilities, apply problem-solving techniques to solving real-world events and apply selected probability distributions to solve problems.

Syllabus

Unit	Topic	Contents	Periods
1	**Probability**	Random phenomena – Basic concepts of Probability – Classical Probability – Empirical Probability – Axiomatic approach to Probability – conditional probability – marginal probability – Independence of events – Bayes theorem.	10
2	**Random Variables**	Events and random variables – Discrete and continuous random variables – Probability distribution – Distribution function – Mathematical expectation – variance – Moment generating and characteristic functions.	12
3	**Discrete Distributions**	Bernoulli, Binomial, Poisson, Geometric distributions, Properties and applications.	12
4	**Continuous Distributions**	Uniform, Cauchy, Normal, Exponential, Gamma distributions – Properties and applications.	12
5	**Limit theorems**	Types of convergence, Chebyshev's inequality, Weak and strong laws of large numbers – De Moivre's – Laplace central Limit Theorem.	10
		Total	**56**

Key Text(s) / Readings

- S.C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand & Sons, 2014.
- J. E. Freund, Mathematical Statistics, Prentice Hall of India., 2013, eighth edition.
- Ross S, A First Course in Probability, Pearson Education., 2013, ninth edition. (Chap. 1-5,7,8)
- J.L. Devore, Probability and Statistics for Engineering and Sciences, Brookes / Cole Publishing Co., 2017, ninth edition.